Akoizo Original Practice - 2027 Rotation Form 4 - Answer key

Division C | Circuit Lab | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) XOR(1,1) equals…

Unit: Digital logic and combinational circuits.
Worked-case data: For A=1, B=1, find half-adder sum and carry.

A. 0

B. 1

C. 2

D. Undefined

Correct: A

2. (2 points) Two 100 Ω resistors in series give…

Unit: Diagrams, nodes, series, parallel, and Ohm's law.
Worked-case data: Find Req for 100 Ω and 300 Ω in parallel on 6 V.

A. 200 Ω

B. 50 Ω

C. 100 Ω

D. 10000 Ω

Correct: A

3. (2 points) An ohmmeter generally requires…

Unit: Breadboards, multimeters, and a first virtual build.
Worked-case data: A known 100 Ω resistor has 2 V across it; an unknown in series has 4 V. Find the unknown.

A. A live mains circuit

B. Maximum possible voltage

C. No instrument instructions

D. A de-energized suitable circuit

Correct: D

4. (2 points) A multiplexer…

Unit: Digital logic and combinational circuits.
Worked-case data: For A=1, B=1, find half-adder sum and carry.

A. Always adds inputs

B. Stores a clocked bit by definition

C. Measures resistance

D. Selects an input

Correct: D

5. (4 points) Why trace nodes first?

Unit: Diagrams, nodes, series, parallel, and Ohm's law.
Worked-case data: Find Req for 100 Ω and 300 Ω in parallel on 6 V.

answer: Topology follows connections, not visual placement. (4 points).

6. (4 points) Why check rail continuity?

Unit: Breadboards, multimeters, and a first virtual build.
Worked-case data: A known 100 Ω resistor has 2 V across it; an unknown in series has 4 V. Find the unknown.

answer: Breadboard rails may be split or connected differently from the assumed layout. (4 points).

7. (4 points) Why test all input combinations?

Unit: Digital logic and combinational circuits.
Worked-case data: For A=1, B=1, find half-adder sum and carry.

answer: Different cases can expose omitted inversions or incorrect branch logic. (4 points).

8. (4 points) Why does divider loading matter?

Unit: Diagrams, nodes, series, parallel, and Ohm's law.
Worked-case data: Find Req for 100 Ω and 300 Ω in parallel on 6 V.

answer: The new branch changes equivalent resistance and the voltage distribution. (4 points).

9. (4 points) Why isolate an unknown for resistance measurement?

Unit: Breadboards, multimeters, and a first virtual build.
Worked-case data: A known 100 Ω resistor has 2 V across it; an unknown in series has 4 V. Find the unknown.

answer: Other paths can alter the reading. (4 points).

10. (4 points) What does a Boolean model omit?

Unit: Digital logic and combinational circuits.
Worked-case data: For A=1, B=1, find half-adder sum and carry.

answer: Propagation delay, thresholds, analog transitions, and noise behavior. (4 points).

11. (2 points) A 10 V ideal battery drives a 5 ohm resistor. Find current.

answer: 2 A (2 points).

Numerical target 2; absolute tolerance 0.004. Accepted units: , A.

12. (2 points) A series pair has resistances 6 and 8 ohms. Find equivalent resistance.

answer: 14 ohms (2 points).

Numerical target 14; absolute tolerance 0.028. Accepted units: , ohms.

13. (2 points) Two identical 12 ohm resistors are placed in parallel. Find equivalent resistance.

answer: 6 ohms (2 points).

Numerical target 6; absolute tolerance 0.012. Accepted units: , ohms.

14. (2 points) An ideal resistor carries 2 A at 10 V. Find power.

answer: 20 W (2 points).

Numerical target 20; absolute tolerance 0.04. Accepted units: , W.

15. (2 points) A node has incoming currents 7 A and 2 A and one outgoing current. Find the outgoing current.

answer: 9 A (2 points).

Numerical target 9; absolute tolerance 0.018. Accepted units: , A.

16. (2 points) Explain why an ideal voltmeter is connected in parallel rather than used as a series current path.

answer: It measures potential difference across a component and has very high input resistance; placing it in series would alter or interrupt the intended current path. (2 points).

17. (2 points) Give the output of an AND gate for inputs 1 and 0, and explain.

answer: 0; an AND output is 1 only when both inputs are 1. (2 points).

18. (2 points) A silicon diode is reverse biased below breakdown. Contrast its ideal current with a forward-biased conducting diode.

answer: In the ideal model the reverse-biased diode blocks current; a forward-biased conducting diode permits conventional current from anode toward cathode. Real leakage and forward voltage require a stated device model. (2 points).

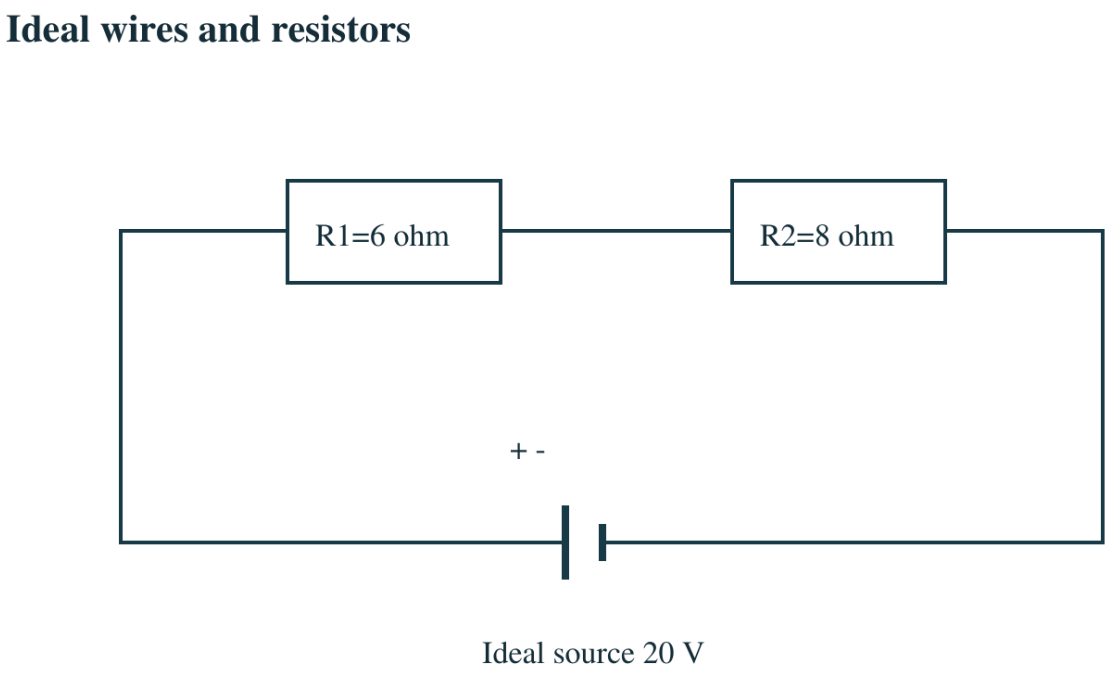
19. (2 points) A closed DC loop has a 9 V source and a 4 V resistor drop. Find the remaining series resistor drop and state the governing law.

answer: 5 V; Kirchhoff voltage law requires the algebraic sum of potential changes around the closed loop to be zero. (2 points).

20. (2 points) Two identical LEDs must be equally bright in a low-voltage teaching circuit. Explain why their branch currents and current-limiting arrangements matter.

answer: Brightness depends on operating current; matched LEDs with suitable branch current limiting and comparable currents can produce comparable brightness. Do not connect them without appropriate current control. (2 points).

21. (2 points) Use Figure F1. Calculate equivalent resistance.



F1. Series resistor circuit (original synthetic teaching figure)

answer: 14 ohms (2 points).

Numerical target 14; absolute tolerance 0.042. Accepted units: , ohms.

22. (2 points) Use Figure F1. Calculate loop current.

Refer to F1. Series resistor circuit (original synthetic teaching figure) shown with the first question in this figure group.

answer: 1.4285714 A (2 points).

Numerical target 1.4285714; absolute tolerance 0.0042857143. Accepted units: , A.

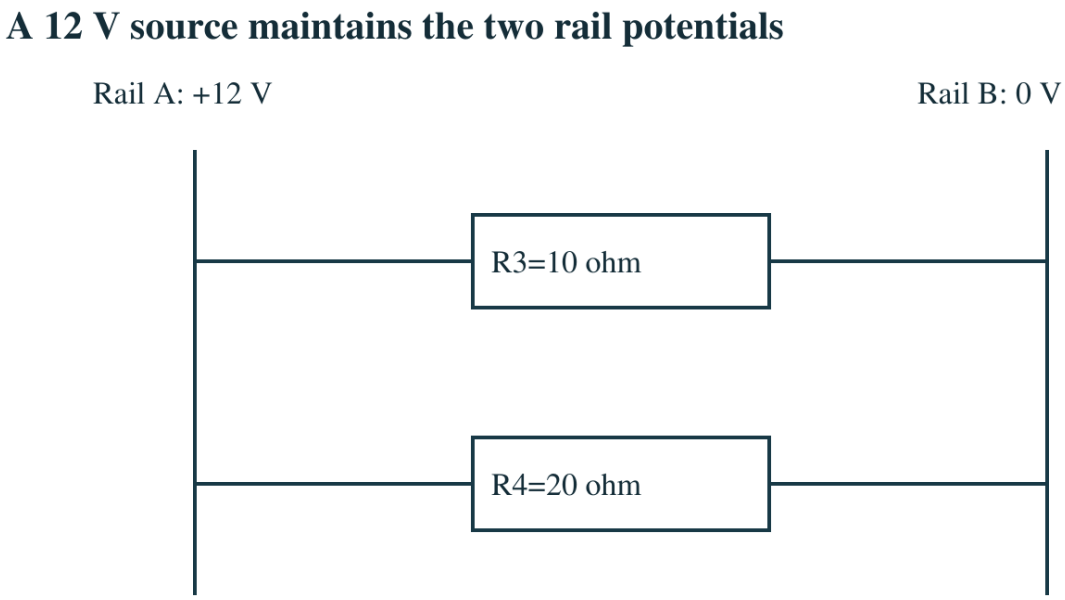
23. (2 points) Use Figure F1. Calculate the potential difference across R1.

Refer to F1. Series resistor circuit (original synthetic teaching figure) shown with the first question in this figure group.

answer: 8.5714286 V (2 points).

Numerical target 8.5714286; absolute tolerance 0.025714286. Accepted units: , V.

24. (2 points) Use Figure F2. Find current through R3.



F2. Two parallel branches (original synthetic teaching figure)

answer: 1.2 A (2 points).

Numerical target 1.2; absolute tolerance 0.0036. Accepted units: , A.

25. (2 points) Use Figure F2. Find current through R4.

Refer to F2. Two parallel branches (original synthetic teaching figure) shown with the first question in this figure group.

answer: 0.6 A (2 points).

Numerical target 0.6; absolute tolerance 0.0018. Accepted units: , A.

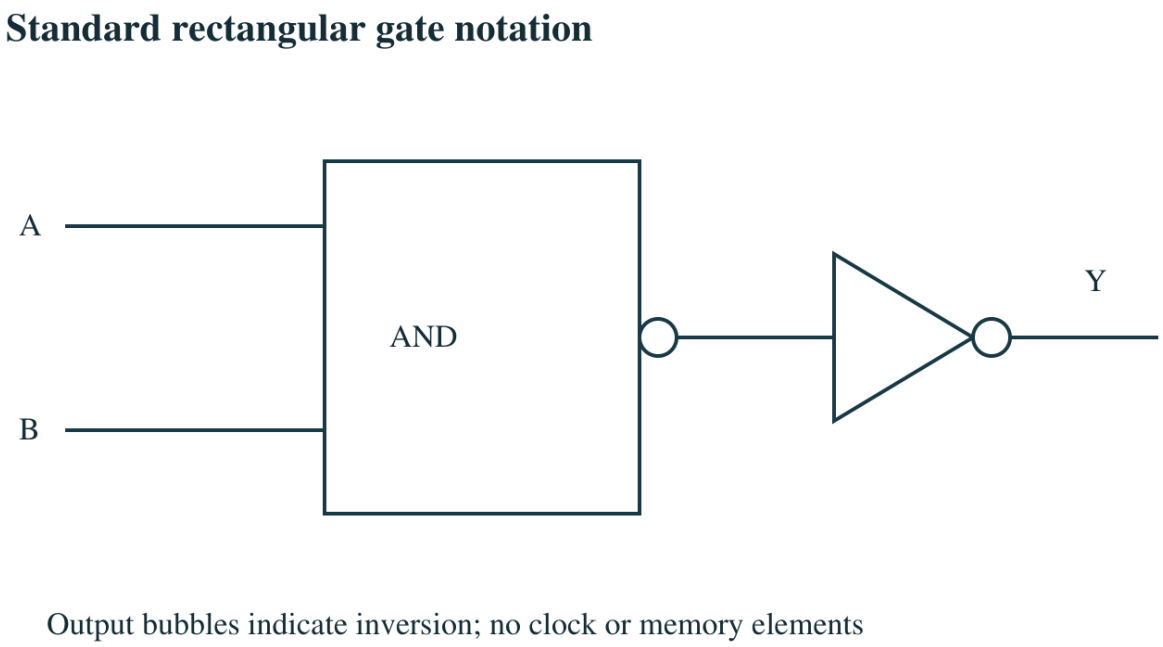
26. (2 points) Use Figure F2. Find equivalent resistance between the rails.

Refer to F2. Two parallel branches (original synthetic teaching figure) shown with the first question in this figure group.

answer: 6.6666667 ohms (2 points).

Numerical target 6.6666667; absolute tolerance 0.02. Accepted units: , ohms.

27. (2 points) Use Figure F3. Find Y for A=1 and B=0.



F3. Combinational logic (original synthetic teaching figure)

answer: 0 (2 points).

Numerical target 0; absolute tolerance 1e-10. Accepted units: .

28. (2 points) Use Figure F3. Find Y for A=1 and B=1.

Refer to F3. Combinational logic (original synthetic teaching figure) shown with the first question in this figure group.

answer: 1 (2 points).

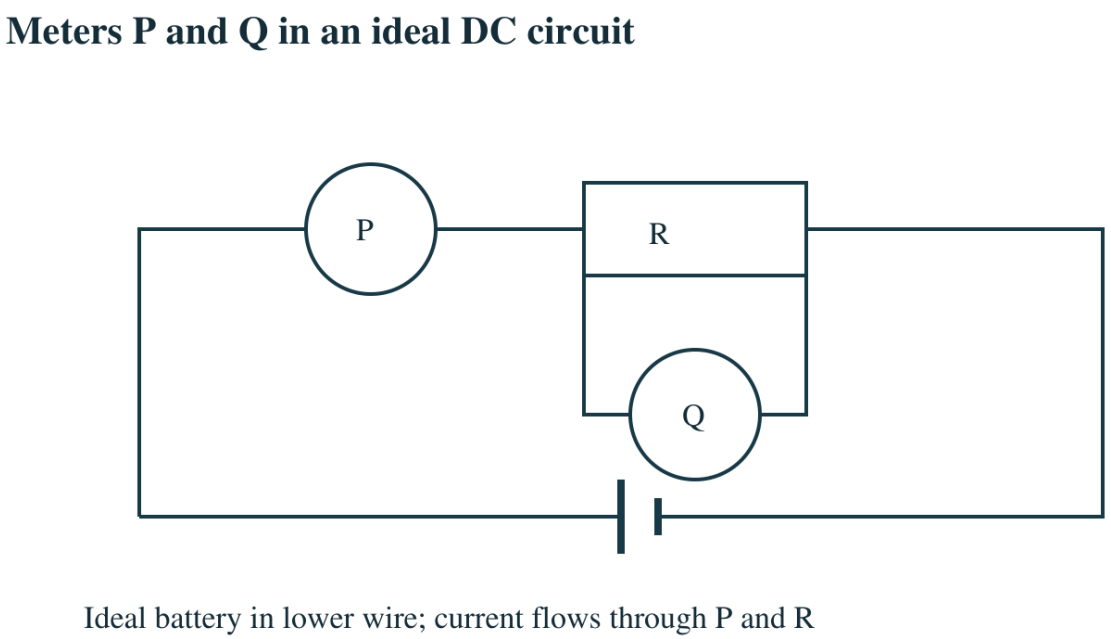
Numerical target 1; absolute tolerance 0.003. Accepted units: .

29. (2 points) Use Figure F3. Name the single gate equivalent to the complete diagram and justify.

Refer to F3. Combinational logic (original synthetic teaching figure) shown with the first question in this figure group.

answer: AND. The NAND result is inverted again, so the two inversions cancel and Y=A AND B. (2 points).

30. (2 points) Use Figure F4. Which meter position is appropriate for a voltmeter? Explain using the wiring.



F4. Measurement connections (original synthetic teaching figure)

answer: Q, because it spans both terminals of R and measures potential difference in parallel. (2 points).

31. (2 points) Use Figure F4. Which meter position is appropriate for an ammeter? Explain.

Refer to F4. Measurement connections (original synthetic teaching figure) shown with the first question in this figure group.

answer: P, because it lies in the series current path through R. (2 points).

32. (2 points) Use Figure F4. Describe the desired input resistance of the meter at Q and its purpose.

Refer to F4. Measurement connections (original synthetic teaching figure) shown with the first question in this figure group.

answer: Very high input resistance, to draw little additional current and reduce loading of the measured circuit. (2 points).